

Distillation and autoregression in video generation

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Why we want to distill T2V models

Even a short video produce very long token sequence

- A 5-second, 16 FPS video at 480x832 already becomes 33k tokens
- Increasing the resolution to 720x1248 raises this to 73k
- Generating the first video with 1.3B model takes 140s

Methods of the distillation T2V models

VideoLCM: Video Latent Consistency Model

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DOLLAR: Few-Step Video Generation via Distillation and Latent Reward Optimization

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miracle.github.io/dollar/

Dual-Expert Consistency Model for Efficient and High-Q

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SF-V: Single Forward Video Generation Model

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Project Page: <https://snap-research.github.io/SF-V>

Motion Consistency Model: Accelerating Video Diffusion with Angled Motion-Appearance Distillation

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<https://yhzhai.github.io/mcm/>

TOWARDS ONE-STEP CAUSAL VIDEO ADVERSARIAL SELF-DISTILLATION

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Project page: <https://github.com/BigAandSmallq/>

Accelerating Video Diffusion Models via Distribution Matching

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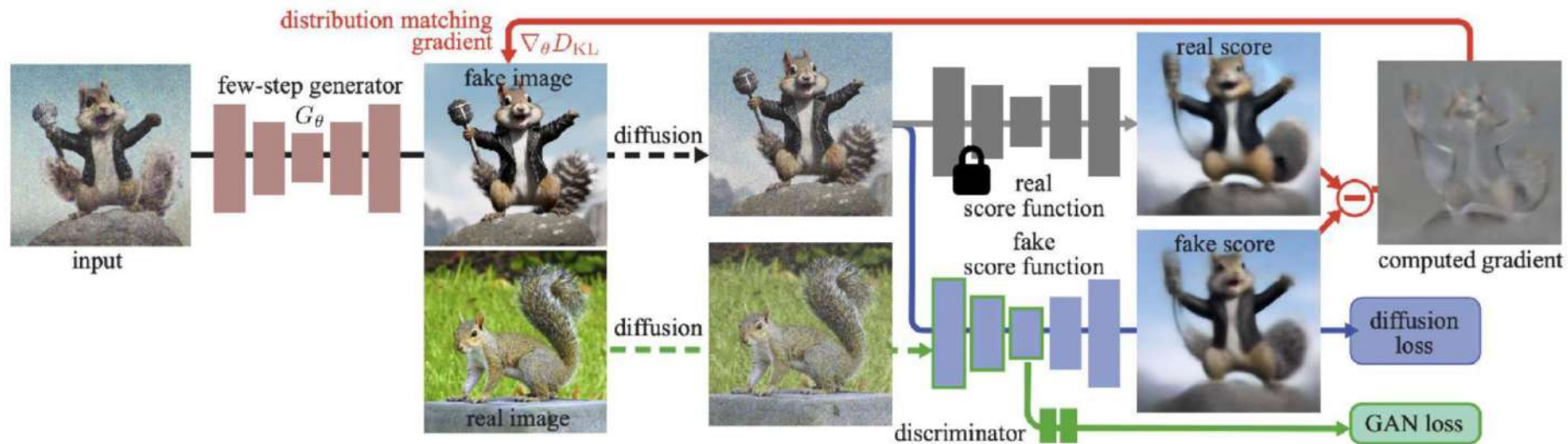
¹Rhymes.AI, ²Nanjing University

Adversarial Post-Training for One-Step Video Generation

Hanchuan Lin¹ Xin Xia¹ Yuxi Ren¹ Ceyu Yang¹ Xuefeng Xiao¹ Lu Jiang¹

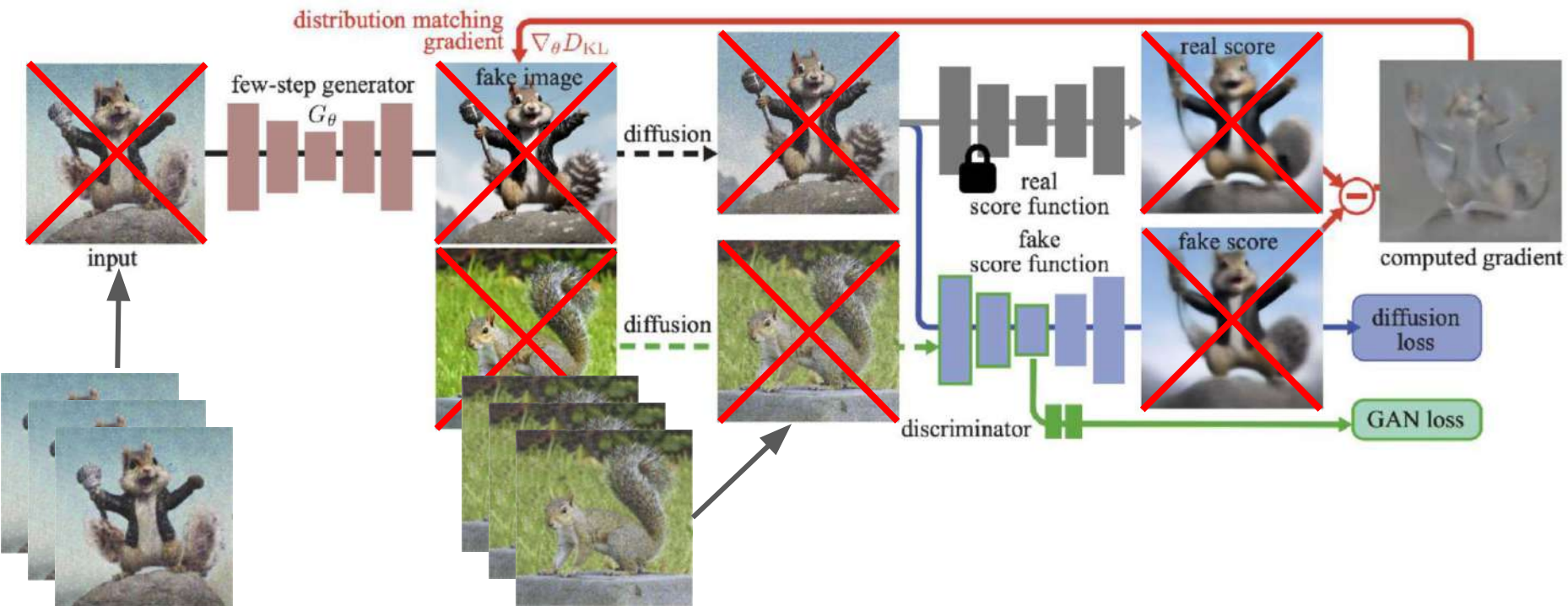
DMD/DMD2 recap

Let's combine GAN and DMD losses



CausVid: DMD2 as is

Just swap image with video



How to generate long videos?

- Fast generation of 5-second videos is already possible
- The next challenge is scaling to longer videos
- In practice we often do not know the desired duration beforehand

There are two main approaches:

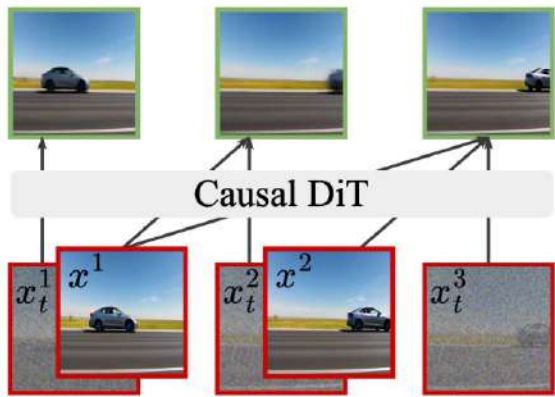
Key frames interpolation
(less commonly used in prior works)

Autoregressive frames generation
(dominant direction)

Key frames interpolation



Autoregressive frames generation | Teacher Forcing

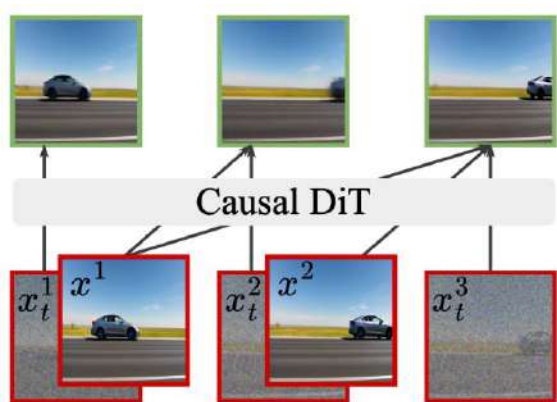


(a) Teacher Forcing Training

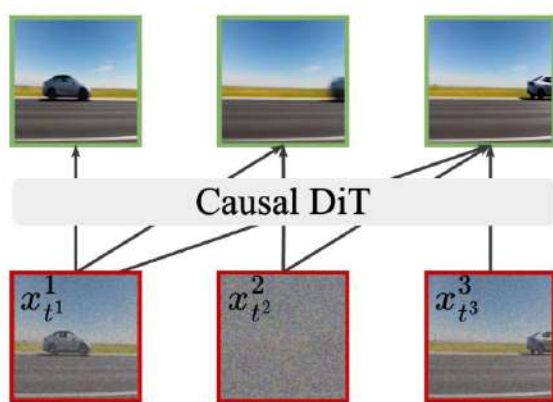
Problems:

- 1) Train-inference mismatch
- 2) Error accumulation

Autoregressive frames generation | Diffusion Forcing



(a) Teacher Forcing Training

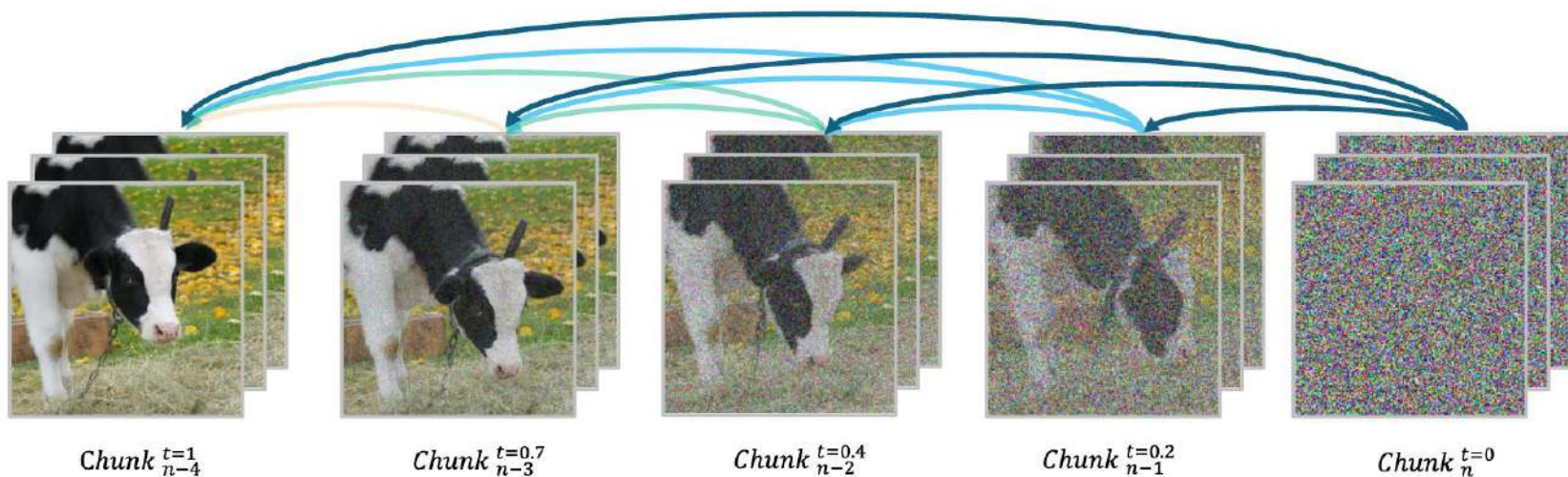


(b) Diffusion Forcing Training

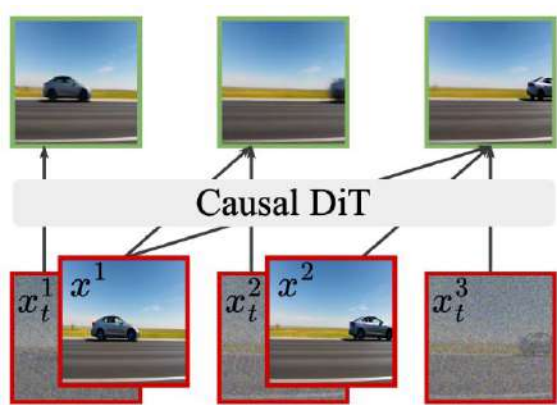
DF is trying to fix the train inference mismatch by conditioning not on clean but on noise frames

Diffusion Forcing Sampling

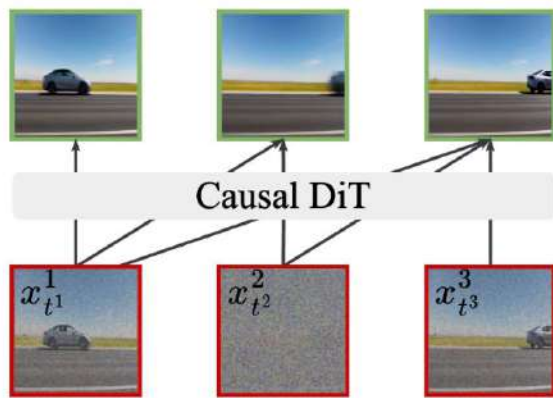
DF accelerates generation by assigning different noise levels to different frames (or chunks) during training.



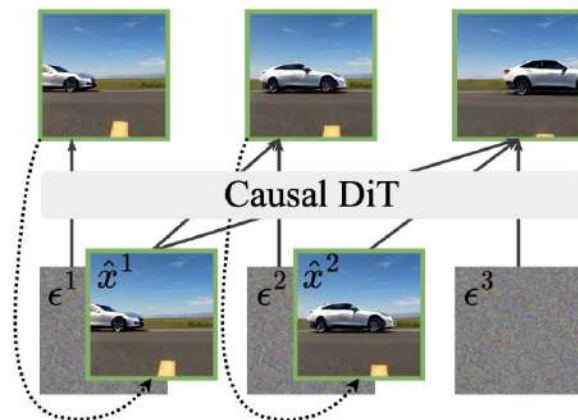
Autoregressive frames generation | Self Forcing



(a) Teacher Forcing Training



(b) Diffusion Forcing Training



(c) Self Forcing Training

SF fix train inference mismatch by conditioning on its own prediction during training.

Self Forcing | Training algorithm

Algorithm 1 Self Forcing Training

Require: Denoise timesteps $\{t_1, \dots, t_T\}$

Require: Number of video frames N

Require: AR diffusion model G_θ (returns KV embeddings via G_θ^{KV})

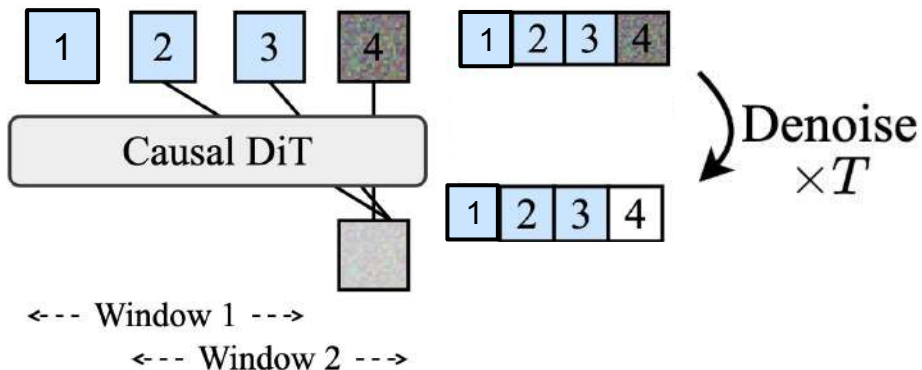
```
1: loop
2:   Initialize model output  $\mathbf{X}_\theta \leftarrow []$ 
3:   Initialize KV cache  $\mathbf{KV} \leftarrow []$ 
4:   Sample  $s \sim \text{Uniform}(1, 2, \dots, T)$ 
5:   for  $i = 1, \dots, N$  do
6:     Initialize  $x_{t_T}^i \sim \mathcal{N}(0, I)$ 
7:     for  $j = T, \dots, s$  do
8:       if  $j = s$  then
9:         Enable gradient computation
10:        Set  $\hat{x}_0^i \leftarrow G_\theta(x_{t_j}^i; t_j, \mathbf{KV})$ 
11:         $\mathbf{X}_\theta.append(\hat{x}_0^i)$ 
12:        Disable gradient computation
13:        Cache  $\mathbf{kv}^i \leftarrow G_\theta^{\text{KV}}(\hat{x}_0^i; 0, \mathbf{KV})$ 
14:         $\mathbf{KV}.append(\mathbf{kv}^i)$ 
15:      else
16:        Disable gradient computation
17:        Set  $\hat{x}_0^i \leftarrow G_\theta(x_{t_j}^i; t_j, \mathbf{KV})$ 
18:        Sample  $\epsilon \sim \mathcal{N}(0, I)$ 
19:        Set  $x_{t_{j-1}}^i \leftarrow \Psi(\hat{x}_0^i, \epsilon, t_{j-1})$ 
20:      end if
21:    end for
22:  end for
23:  Update  $\theta$  via distribution matching loss
24: end loop
```

Because the student model is causal, KV cache is available, allowing to reuse keys and values in attention.

After generating the output, we run one extra forward at $t=0$ to refresh the KV cache.

Self Forcing | Long video generation via rolling KV cache

conditioning on ALL previous frames (keep full KV cache)



conditioning only on L last frames (evict oldest KV cache)

